An Explainable Deep Learning Framework for Resilient Intrusion Detection in IoT-Enabled Transportation Networks

Ayodeji Oseni, Nour Moustafa, *Senior Member, IEEE*, Gideon Creech, Nasrin Sohrabi, Andrew Strelzoff, Zahir Tari, and Igor Linkov

***Abstract*— The security of safety-critical IoT systems, such as the Internet of Vehicles (IoV), has a great interest, focusing on using Intrusion Detection Systems (IDS) to recognise cyber-attacks in IoT networks. Deep learning methods are commonly used for the anomaly detection engines of many IDSs because of their ability to learn from heterogeneous data. However, while this type of machine learning model produces high false-positive rates and the reasons behind its predictions are not easily understood, even by experts. The ability to understand or comprehend the reasoning behind the decision of an IDS to block a particular packet helps cybersecurity experts validate the system's effectiveness and develop more cyber-resilient systems. This paper proposes an explainable deep learning-based intrusion detection framework that helps improve the transparency and resiliency of DL-based IDS in IoT networks. The framework employs a SHapley Additive exPlanations (SHAP) mechanism to interpret decisions made by deep learning-based IDS to experts who rely on the decisions to ensure IoT networks' security and design more cyber-resilient systems. The proposed framework was validated using the ToN_IoT dataset and compared with other compelling techniques. The experimental results have revealed the high performance of the proposed framework with a 99.15% accuracy and a 98.83% F1 score, illustrating its capability to protect IoV networks against sophisticated cyber-attacks.**

***Index Terms*— Explainable AI, network intrusion detection, deep learning, IoT, security, Internet of Vehicles (IoV).**

I. Introduction

IN RECENT years, the Internet of Things (IoT) has emerged as a disruptive technology enabling devices armed with sensors and networking capabilities to connect to the Internet. IoT devices, such as smart gadgets, appliances, machines, and buildings, collect, process and transmit data through low-bandwidth communication channels and sometimes operate within complex systems called IoT ecosystems. A recent study from Juniper Research indicates that the total number of IoT connections will reach 84 billion by 2024, which is a significant rise from 35 billion in 2020 [1]. With the record growth in the application of IoT in health monitoring systems, autonomous vehicles, robotic systems, industrial control systems and automatic pilot avionics, the market revenue from IoT in 2020 exceeded $212 billion worldwide. Smart cars and other connected vehicles, which together with smart lights and sensors, form the Internet of Vehicles (IoV), represent a significant portion of the current IoT market [2]. For many IoT systems, security is often an afterthought making them open and vulnerable to several cyber-attack scenarios as well as increasing the potential of harmful and dangerous usage [3]. For instance, an attack against critical infrastructures, such as those used in electricity generation, water treatment and smart cities, may result in unintended consequences such as fire, explosion or even the loss of human life. Furthermore, with the increased use of IoT devices in safety-critical applications such as smart city services, smart grids, and smart manufacturing, a question that is often raised is how best to ensure the security and safety of the devices.

With the increased use of information and communication technologies in transportation systems to improve road safety, traffic monitoring and passenger comfort, the security of the systems and their operations within the Internet of Vehicles (IoV) ecosystem is now more important than ever before [4]. While there has been some research focus on detecting anomalous events on IoV networks using AI techniques [5], the lack of transparency and cyber-resiliency of some of the proposed methods is concerning to many cyber security experts. According to [6], existing security technologies used for securing traditional computer networks are often not suitable for many safety-critical IoT and IoV networks that have stringent security and safety requirements. Also, with many consumer-based IoT devices operating within resource-constrained communication networks [7], existing security solutions used in traditional networks, such as firewalls and intrusion detection systems, cannot be directly

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Ayodeji Oseni and Nour Moustafa are with the School of Engineering and Information Technology, University of New South Wales Canberra, Campbell, ACT 2612, Australia (e-mail: a.oseni@student.unsw.edu.au; nour.moustafa@unsw.edu.au).

Gideon Creech is with Volos Services Pty Ltd., Wamboin, NSW 2620, Australia (e-mail: zero@volos.services).

Nasrin Sohrabi and Zahir Tari are with the Centre of Cyber Security Research and Innovation (CCSRI), School of Computing Technologies, RMIT University, Melbourne, VIC 3000, Australia (e-mail: nasrin.sohrabi@rmit.edu.au; zahir.tari@rmit.edu.au).

Andrew Strelzoff and Igor Linkov are with the U.S. Army Engineer Research and Development Center, Vicksburg, MS 39180 USA (e-mail: andrew.p.strelzoff@usace.army.mil; igor.linkov@usace.army.mil).

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applied to IoT applications due to their globally accessible nature and resource-constrained characteristics [8].

Many studies have concentrated on improving the cyber-resilience of transportation systems to meet their stringent security and safety requirements [9]. Intrusion detection is considered a powerful means of achieving cyber-resilience in IoV networks. Cyber-resilience is the ability of a system to absorb disruption or attack, recover to a pre-disruption functionality, and adapt its protocol and configuration to become more resilient to known threats. It is an important concept that has recently attracted attention in several IoT domains such as intelligent transportation systems [10]. To maintain the confidentiality, integrity, availability and safety of critical IoT systems, they need to be resilient against cyber and physical attacks. However, the current state of those attacks on critical IoT infrastructure requires systems that absorb or prevent attacks and address key requirements such as quick recovery and adaptation.

Motivation – Recent works on intrusion detection systems have focused on the use of deep learning models due to their ability to learn from heterogeneous data automatically [11]. The idea that deep learning models are inherently not transparent is one of the reasons for the cautious adoption of deep learning in many safety-critical IoT systems, such as those used in health monitoring and autonomous driving [12]. Understanding the reasoning behind a learning system's decision helps foster trust and allows experts to verify that the system is safely and securely solving a problem. In addition, due to the complexity and heterogeneity of IoT-enabled transportation networks, DL-based intrusion detection systems with explainable capability can provide transparency into how cyber-attacks are predicted [5]. Based on this information, cybersecurity experts are able to design intrusion detection systems that can better absorb attacks and adapt their protocols and configurations to be more resilient to complex attacks. Therefore, the primary focus of this research is to improve the transparency and resilience of deep learning-based intrusion detection systems in IoV networks.

Contribution – The main contributions of this paper are as follows:

- We propose a framework for the local and global explanation of AI-based intrusion detection systems in the context of IoT and IoV communication networks. This work constitutes a significant contribution to literature as it helps cybersecurity experts better understand the output of the AI models used for intrusion detection in IoT networks.
- We present an application of the Deep SHAP method for explaining the output of Intrusion Detection Systems in IoT networks. While other SHAP methods have been used to provide an explanation for AI models in various computer vision and natural language processing contexts, our work provides the first-ever application of Deep SHAP in Network Intrusion Detection Systems.
- We evaluate and test the proposed method using a CNN-based Intrusion Detection System with the ToN_IoT dataset for potential implementation in IoV networks.

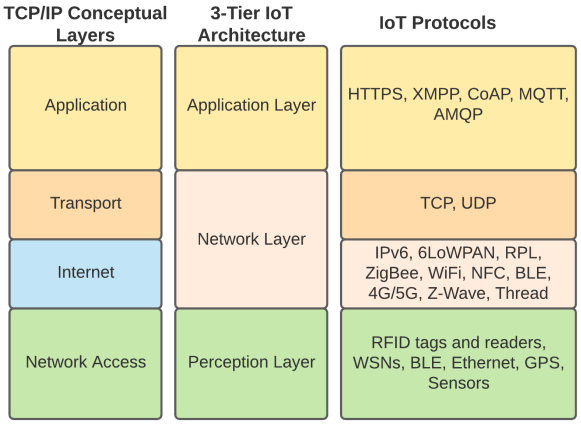


Fig. 1. Three-tier IoT architecture and protocols.

The rest of this paper is organised as follows. Section II discusses the background and related work of this work. The proposed framework follows this in Section III. In Section IV, the experiments and results are described. Finally, Section V presents the conclusion and future work of this study.

II. Extant Literature

This work is related to the explainability of AI-based intrusion detection systems for IoT networks. This section provides foundational concepts of IoT architectures and protocols, IoT assets and applications, and some security challenges in IoT. We then give a brief background of Intrusion Detection Systems (IDS) and a review of previous works on intrusion detection systems in IoT networks.

A. IoT Architectures and Protocols

IoT is one of the core technologies supporting smart services such as smart homes, smart cities and smart grids. It enables inter-connectivity among various 'smart' devices, which are often embedded with various heterogeneous components such as sensors, actuators, processors and transceivers [13]. While there is no consensus on a standard IoT architecture, many studies have proposed different architectures for describing IoT. The most basic architecture is the three-layer IoT architecture shown in Fig. 1 comprises an application layer, a network layer and a perception layer. The three-layer architecture is sufficient to describe the core functional blocks of an IoT system, where each layer performs a specific set of services for the three layers.

The *perception layer*, also known as the physical layer, is the last layer of the IoT architecture (Fig.1) and comprises components such as sensors and actuators. Sensors collect information about the environment while actuators perform specific functions in response to system instructions. The *network layer* is considered the backbone of IoT systems as it enables connectivity with other smart devices, network devices, computing nodes and the Internet [14]. It is responsible for transmitting data received from sensors to other nodes for processing. Connectivity among IoT nodes can be wired or wireless. However, wireless connectivity is more common in

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TABLE I
COMMON ATTACKS IN IOT

IoT Conceptual Layer	Attack Type
Application Layer	Password attack, Malware attack, Ransomware attack, Cross-site Scripting (XSS) attack, Backdoor attack
Network Layer	Replay attack, Man-in-the-middle attack, Sinkhole attack, Scanning attack, Denial/Distributed Denial of Service (DoS/DDoS) attack, Wormhole attack, Routing information attack
Perception Layer	Tampering attack, Malicious Code Injection attack, RF Interference or Jamming attack, Side Channel attack, Fake Node Injection attack

many IoT ecosystems. The system designer often determines network layer communication protocols used in IoT. The *application layer* directly interfaces with users. Its primary function is to provide application-specific services to the users. It provides a usability interface for the data collected at the perception layer. The usability interface can be via computer applications with GUI, mobile applications, websites or cloud services [15].

B. IoT Assets and Their Security Challenges

IoT assets are defined as the hardware, software and data that operate at any of the IoT layers. Typical IoT assets include smart devices, communication protocols, communication channels, and applications. These assets are usually the target of attackers seeking to breach the security of an IoT system. [16]. It is vital to protect autonomous vehicles, traffic lights, dynamic traffic management systems, and parking guidance systems against complex cyber-attacks in smart transportation. Cyber-attacks that exploit smart transportation include DoS/DDoS, spoofing, backdoor, man-in-the-middle and false data injection.

Security is one of the main challenges to deploying IoT systems and ongoing operations. Unfortunately, with security often being an afterthought for many IoT devices and systems, attacks on IoT infrastructure and networks often originate from simple security problems. In particular, autonomous vehicles have been found to be vulnerable to attacks that aim to hijack vehicles while in motion, thereby posing severe risks to drivers and other road users. Likewise, a cyber-attack on safety-critical IoT applications, such as smart grids and smart manufacturing, would potentially devastate the systems and humans. In Table I, we broadly classify common attacks in IoT into three categories corresponding to the three-layer IoT architecture. Perception layer attacks are typically launched by attackers that are within close proximity to IoT networks or devices [17], while network layer attacks can be launched without being physically close to the network or device. On the other hand, application-layer attacks typically target the IoT application.

1) Threat Modeling in IoT: In the context of IoT, threat modeling helps to proactively avoid introducing vulnerabilities to a system while in design and development life-cycle phases [18]. As a continuous process used to enhance security in many systems and networks, threat modeling also helps identify vulnerabilities in existing IoT systems and networks. In this study, we assess the risks of an attack on IoT networks based on the following threat modeling components:

*a) **Assets**:* IoT assets are defined as the hardware, software and data that operate at any of the layers described in Section II-A. Typical IoT assets include smart devices, communication protocols, communication channels, and applications. These assets are usually the target of attackers seeking to breach the security of an IoT system [16].

*b) **Threats and Vulnerabilities**:* A critical component of threat modeling in IoT is the identification of threats or attacks that a system is exposed to, as well as vulnerabilities in the system. While various attacks on IoT systems can occur at any layer of the IoT architecture, the majority of the attacks target the network layer. In Table II, we provide a brief description of the attack types we consider in this study.

*c) **Mitigation Strategies**:* These refer to various techniques that can be used to limit the impact of a threat against IoT assets. A number of mitigation strategies can be used to prevent or limit the impact of an attack on IoT and IoV assets; however, in this work, we focus on the use of intrusion detection systems in detecting and preventing attacks on IoT networks.

C. Intrusion Detection Systems

An IDS monitors a host or network for any malicious or abnormal behaviours that violate security. In IoT architectures, an IDS operates within the network layer. Intrusion detection systems typically use one of two detection approaches, namely: signature-based detection, anomaly-based detection or a hybrid of both [19].

1) Signature-Based IDS: monitors network traffic and compares packet sequences and patterns to known attack signatures to flag any suspicious behaviour. The signature-based detection approach is effective against known attacks with low false positives rates. However, this approach requires the IDS to store attack signatures and does not detect zero-day attacks. Snort is a common open-source, signature-based intrusion detection and prevention system which can detect and prevent attacks based on pre-defined rules [20]. The benefit of Snort over other signature-based IDSs is that it can analyze network traffic in real-time while also discarding packets that degrade network performance.

2) Anomaly-Based IDS: baselines normal network traffic and then flags any deviation as an anomaly. This detection approach is effective against zero-day and modern attack strategies, but they often have high false-positive rates. Different techniques can be used for anomaly detection, with the most common being statistical-based, machine learning, and deep learning techniques.

3) Hybrid IDS: A hybrid of signature-based and anomaly-based detection is sometimes used in intrusion detection sys-

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TABLE II
DESCRIPTION OF COMMON CYBERATTACKS IN IOT NETWORKS

Attack Type	Description
Man-in-the-middle (MITM)	This is an attack where an adversary actively intercepts communications between two parties with the goal of eavesdropping or modifying the traffic path.
Distributed Denial of Service (DDoS)	DDoS attack is a malicious attempt to disrupt the normal traffic of a targeted system or network by overwhelming the target with more traffic than it can accommodate. This attack is often originated from multiple sources.
Denial of Service (DoS)	A DoS attack is one in which the attacker aims to render the target system or network unavailable for its intended use.
Scanning	Port scanning attack is used by an adversary to discover weak points or vulnerabilities in a network. It usually involves huge amounts of requests to different ports or IP Addresses within a short period of time.
Cross-site Scripting (XSS)	The Cross-Site Scripting (XSS) attack is a type of injection attack in which an adversary injects malicious codes to trusted websites, enabling the adversary to compromise the interactions that users have with the website.
Injection	Injection attack is a type of attack that enables an adversary to inject code or malware onto a system in order to execute remote commands that can give the attacker control of the system.
Password	A password attack is simply an attack where an adversary make attempts to steal or compromise a password.
Ransomware	A ransomware attack is one in which an adversary introduces a malware into a system to encrypt critical data on the system.
Backdoor	A backdoor attack is one in which an adversary uses a specific type of malware to evade authentication to a target system.

tems to take advantage of detecting known and zero-day attacks with relatively low false-positive rates.

[9] developed a state-of-art knowledge-driven IDS, Kalis, which provides real-detection capabilities to many IoT communication protocols and applications across a wide range of IoT systems. Kalis is a hybrid of signature and anomaly-based IDS and automatically adapts its detection techniques based on collected network features [21]. Compared to SVELTE, which is limited to devices communicating via the RPL protocol, Kalis can be applied to diverse protocols and communication mediums. Kalis was implemented using Java and has been noted to have a high detection rate in different attack scenarios, including DoS attacks on IoT [9]. In its current state, Kalis did not use any machine learning technique and relies on stored information about the features of monitored entities and networks to detect malicious traffic.

Many modern IoT ecosystems are driven by technologies such as cloud, big data and artificial intelligence. More specifically, IoT devices used in many environments with high security and safety requirements are often supplemented by cloud storage, big data analytics and data mining [21]. These technologies have led to the evolution of security attacks, with attackers constantly developing highly sophisticated and novel tactics, techniques and procedures (TTP) that evades traditional IDSs, especially signature-based solutions. In recent years, researchers have been investigating the use of machine learning and deep learning techniques for detecting security threats in safety-critical IoT networks. For instance Abeshu and Chilamkurti [22] proposed a novel deep learning scheme for detecting attacks on distributed IoT networks supported by fog computing nodes. The solution employs distributed deep learning, through parameters and model exchange between worker and master nodes, to detect attacks on IoT/fog networks. Thamilarasu and Chawla evaluated the performance of the intelligent IDS using network traffic generated from a simulated IoT testbed created using the Texas Instruments sensor tags CC2650.

In the context of IoV networks, Kumar *et al.* [24] proposed a framework, P2SF-IoV, that provides security and privacy for IoV networks based on blockchain and deep learning techniques. The proposed framework comprises a blockchain module that secures data transmission in the network and a deep learning module that detects intrusions. The deep learning module utilises a stacked Long-Short Term Memory (SLSTM) model to detect and identify cyber-attacks in IoV networks. Kumar *et al.* demonstrated through experimental analysis that their DL-based SLSTM model trained on the ToN-IoT and IoT-Botnet datasets achieved high multi-class detection rates for most of the attacks and normal classes present in both datasets. In related work, Zhang *et al.* [25] proposed a DNN-based IDS that combines gradient descent with momentum (GDM) and gradient descent with momentum and adaptive gain (GDM/AG) to improve the overall efficiency and accuracy of intrusion detection systems in intelligent vehicles. They evaluated the performance of their model using data collected from a Control Area Network (CAN), and experimental results show that their proposed IDS achieves higher detection accuracy compared to some earlier works.

D. Explainable AI

With the growing popularity of deep learning models in safety-critical IoT and IoV systems, there is an increasing demand for explanations of their predictions [26]. Explainable AI (XAI) provides explanations about its actions in a systematic and interpretable manner that human users can

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understand [27]. As defined by Gunning [26], "XAI creates a suite of AI techniques that enables human users to understand, appropriately trust, and effectively manage the emerging generation of artificial intelligent partners". The details or reasons to explain a model, its input or prediction, depends upon the audience or recipient of the explanation [27]. While much literature use 'interpretability' and 'explainability' interchangeably, the two terms are different. Explainability is an active characteristic of a model that describes its steps to expose its often complex internal functions to humans. On the other hand, interpretability is a passive property of a model that refers to the level at which the mechanisms of the model makes sense to humans.

Many 'shallow' machine learning models, such as Naive Bayes, Linear regression, Logistic regression, and Decision trees, are intrinsically interpretable on a modular level [28]. While providing far superior predictive ability than 'shallow' machine learning models, deep learning models generally do not make predictions. For users and stakeholders, understanding the reasoning behind a model's predictions helps foster trust and verify that the model is solving a problem robustly and securely. The in-transparency of the "black-box" deep learning models is one reason for their cautious adoption in many safety-critical domains. To help users interpret the predictions of black-box models, researchers have been investigating several post-hoc explainability techniques. Some of the post-hoc explainability techniques (portrayed in Fig 2) used to convert deep learning models into explainable models include:

- *Text Explanations*: This approach is used to provide text data that helps to explain the reasoning behind a model's predictions.
- *Feature Relevance Explanations*: This approach is used to explain the complex internal functions of the model through the computation of relevance score for the model's managed variables.
- *Local Explanations*: This technique is used to construct explanations by measuring a model's response to local changes.
- *Explanations using representative examples*: This approach is used to provide a better understanding of the training dataset and how it influences a model's decision.
- *Explanations using surrogate functions*: This technique is used to create an intrinsic or white-box model that simplifies the original model's internal functions. The new, simplified model is often more transparent and interpretable while maintaining a similar performance score as the original "black-box" model [27].
- *Visual Explanations*: The visual explanation technique provides visualization of the model's behaviour. It is used in image classification tasks to generate a caption of why an image is predicted to belong in a particular class.

The local explanations and feature relevance explanation methods are the most adopted for explaining deep neural networks [27]. The feature relevance explanation technique explains the internal functioning of a 'black-box' model by computing the relevance score for each of its input features. Approaches that have been proposed for feature relevance explanations include SHapley Additive exPlanations (SHAP) [29], real-time saliency method, influence functions and sensitivity analysis [30].

In recent years, some works on improving the transparency of learning models for intrusion detection systems have emerged. For instance, Amarasinghe and Manic [31] proposed a methodology for providing offline and online feedback to users about the predictions of DNN-based IDS before and during deployment. In their work, the authors used the Layer-wise Relevance Propagation (LRP) method to estimate user feedback in the form of the contribution of each input feature to the model's prediction. Their system was implemented using the KDD-NSL dataset for only two attack types (DoS and Probe). The LRP method used in their work relies on a Taylor series close to the prediction point rather than a partial derivative at the prediction point itself. Li *et al.* [32] also proposed an approach for explaining the predictions of anomaly-based IDS through local non-linear approximation of the decision boundary of the IDS model. While they successfully demonstrated the feasibility of their approach by explaining predictions of an RNN-based IDS model, the LEMNA explanation method [33] used in their work does not effectively handle the dependency among multiple input samples.

In this work, we proposed using the SHAP framework to explain the predictions of DL-based intrusion detection systems in IoT networks. According to the authors [29], SHAP unifies six existing methods and shows improved computational performance and better consistency with human intuition than previous methods. Thus, we provide a brief overview of this framework in section II-E.

E. SHapley Additive exPlanations (SHAP) Framework

SHAP is a feature relevance explanation framework proposed by Lundberg and Lee [29] which assigns a feature importance value for each particular prediction. It is based on the mathematical foundation of Shapley values from cooperative game theory. SHAP provides explainability by identifying a class of additive feature attribution methods and then showing the presence of a unique solution in this class based on a set of three desirable properties, namely, local accuracy, missingness and consistency.

For a given set of input x and model $f(x)$, the goal of SHAP is to explain the prediction of input x by calculating the contribution of each feature to the prediction. In their work, Lundberg and Lee [29] represented the Shapley value explanation as an additive feature attribution method. The additive feature attribution method defines a set of simplified local inputs x' that maps to the original input through a mapping function $x = h_x(x')$ and explanation model $g(x')$:

$$g(x') = \phi_0 + \sum_{i=1}^{N} \phi_i x'_i \tag{1}$$

where ϕ_0 is the null output of the model and ϕ_i is the explained effect of feature i, which is referred to as its attribution.

According to the authors, SHAP is currently the only post-hoc ML explanation method that satisfies all three desirable properties of *local accuracy*, *missingness* and *consistency*.

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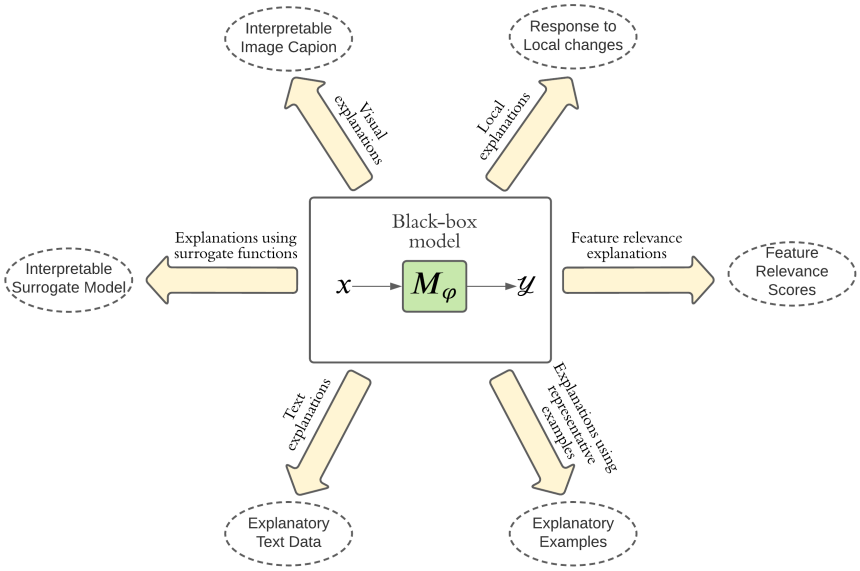


Fig. 2. Different post-hoc explainability approaches for deep learning models.

The local accuracy property states that the input x and simplified input x' are approximately the same. The actual model $f(x)$ and explanation model $g(x')$ should produce similar outputs. The missingness property states that if a feature is missing from a model's original input, its attribution ϕ_i has no impact. The consistency property states that if the original model changes such that a particular feature's contribution also changes, then the attribution should not change in the opposite direction [29].

How SHAP values work: SHAP values interpret how a given feature value increases or decreases a model's prediction. So, for a model that predicts whether a network traffic flow is normal or an attack, SHAP enables us to know the extent to which a feature drove a given prediction, say '*flow duration*', being a certain value, instead of some baseline value. SHAP values have been found to significantly increase the transparency of black-box models as they provide both local and global explanations [34]. Lundberg and Lee [29] acknowledged that calculating SHAP values is challenging and therefore proposed some approximation algorithms by drawing insights from existing additive feature attribution methods. *Kernel SHAP* is a model-agnostic approximation method for computing SHAP values described by Lundberg and Lee [29]. Model-specific approximation methods for computing SHAP values also proposed by Lundberg and Leeincludes *Linear SHAP, Low-Order SHAP, Deep SHAP* and *Max SHAP*.

Amongst the different approximation algorithms for computing SHAP values, Deep SHAP was specifically developed to approximate SHAP values for deep learning models. It builds on an existing connection between DeepLIFT, a recursive prediction explanation method for deep neural networks [35], and Shapley values to efficiently compute SHAP values for deep neural networks. We, therefore, start by providing a brief overview of DeepLIFT and Shapley values before delving into the theoretical foundations of Deep SHAP.

1) DeepLIFT: DeepLIFT is a recursive prediction explanation algorithm that relies on the concept of a difference from a reference activation to compute importance scores in neural networks. A reference activation is an activation that a neuron has when presented with a reference input. Formally, for a target neuron t with difference from reference Δt, DeepLIFT attributes a contribution $C_{\Delta x_i \Delta t}$ to each input x_i such that:

$$\sum_{i=1}^{n} C_{\Delta x_i \Delta t} = \Delta t, \tag{2}$$

where $t = f(x)$ is the neural network output, $\Delta t = f(x) - f(r)$, $\Delta x_i = x_i - r_i$, and r is the reference input.

2) Shapley Values: The Shapley value is a method proposed by Shapley [36] for fair distribution of "payouts" for contributions made by players in a game. This method has been adapted for explaining ML and DL models where the output of a model can be explained by assuming that the "game" is a prediction task and the features of an input instance represent the "players". The Shapley value then represents the average minimum contribution of each feature value to the model output across all possible coalitions [28]. Mathematically, the Shapley Values are computed as:

$$\phi_i = \sum_{S \subseteq \{x_1,\ldots,x_n\} \setminus \{x_i\}} \frac{|S|!(N - |S| - 1)!}{N!} \times [f(S \cup x_i) - f(S)] \tag{3}$$

where x is the vector of feature values, $S \subseteq N$ is a subset of features, and N is the set of all features used in the model. $f(S)$ is the prediction for feature values in set S.

It is computationally intensive to calculate exact Shapley values as this would require all possible coalitions (sets) of feature values to be evaluated with and without the i-th feature. A proposed solution for this problem is to sample coalitions of features and then limit the number of iterations, reducing the

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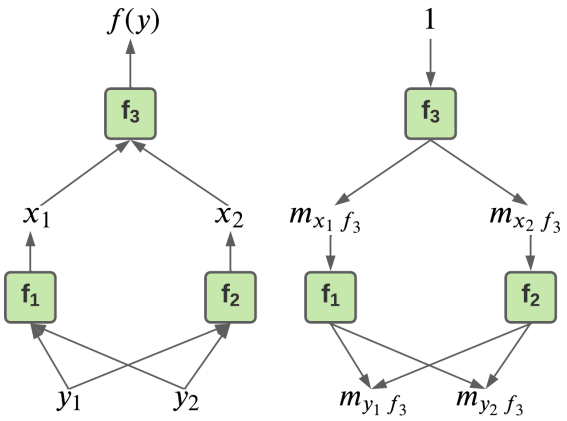


Fig. 3. Compositional models such as deep neural networks are comprised of multiple simple components.

computation time but increasing the variance of the obtained Shapley values. [28].

3) Deep SHAP: Deep SHAP is a fast model-specific approximation approach for explaining the predictions of deep neural networks [29]. This technique was derived from an existing connection between DeepLIFT and Shapley values. Deep SHAP benefits from DeepLIFT's style of back-propagation that makes it possible to compute fast approximations of Shapley values for a deep neural network. Deep SHAP combines the SHAP values calculated for smaller parts of a neural network into SHAP values for an entire network through the back-propagation of DeepLIFT's multipliers. Formally, we define DeepLIFT's multiplier as the contribution of Δx to Δt divided by Δx:

$$m_{\Delta x \Delta t} = \frac{C_{\Delta x \Delta t}}{\Delta x} \tag{4}$$

where $\Delta x = x - r$ and r is the reference input.

According to Lundberg and Lee [29], SHAP values for components of deep neural networks, such as in Fig. 3, can be efficiently solved analytically using a composition rule that allows fast approximation of values for the whole network.

If we represent SHAP value $\phi_i = C_{\Delta x \Delta t}$ and reference input r as $E[x]$,

$$m_{x_j f_3} = \frac{\phi_i(f_3, x)}{x_j - E[x_j]} \tag{5}$$

$$\forall_{j \in 1,2} m_{y_i f_j} = \frac{\phi_i(f_j, y)}{y_i - E[x_i]} \tag{6}$$

$$\text{using chain rule, } m_{y_i f_3} = \sum_{j=1}^{2} m_{y_i f_j} m_{x_j f_3} \tag{7}$$

$$\phi_i(f_3, x) \approx m_{y_i f_3}(y_i - E[x_i]) \tag{8}$$

III. Proposed Framework

In this section, we propose a framework for explaining the predictions of deep learning-based intrusion detection systems in IoT/IoV networks using the Deep SHAP algorithm discussed in Section II-E.3. The proposed framework aims to provide insights into how decisions made by DL-based network intrusion detection systems (NIDS) can be explained to cybersecurity experts that rely on these decisions to ensure the security of IoT systems and design more cyber-resilient IoT systems. In particular, the proposed framework can be used to discover and explain cyber-attacks like DoS and DDoS in IoV networks. This framework shown in Fig. 4 depends on generative deep learning that learns multivariate and time-series data collected from IoV networks and can distinguish legitimate and suspicious instances based on the most representative features. The framework comprises two main components; a deep learning-based network intrusion detection system (NIDS) and an Explanation unit. The novelty of our work is based on the explanation unit, which utilizes the Deep SHAP algorithm to provide local and global explanations for the predictions made by the intrusion detection system in IoT/IoV networks. In comparison with similar frameworks proposed for security and privacy in IoV networks [24], [25], our framework incorporates an explainable unit that can be used to improve the transparency of any IDS. Furthermore, insights derived from the local and global explanations can be used to enhance the cyber-resiliency of IoV systems.

A. Explanation Model

The explanation model is based on the Deep SHAP algorithm described in Section II-E.3. The steps involved in implementing the Deep SHAP algorithm is described as follows:

- Select a set of background examples from training dataset
- Estimate explainer object for a differentiable DL model using the background dataset
- Compute approximate SHAP values for the model applied to test data 'X'

The background examples are integrated over for each sample; hence to reduce computational complexity, only about 100 or 1000 random background samples should be used, not the entire training dataset. By integrating over the background samples, Deep SHAP computes approximate SHAP values such that

$$\sum \phi_i = f(x) - E[f(x)] \tag{9}$$

where ϕ_i is the sum of SHAP values for all features, $f(x)$ is the current model output and $E[f(x)]$ is the expected model output passed on the background samples.

B. Local Explanation

Local explanation, also known as the explanation of individual predictions, involves the interpretation of a model's output based on a single data instance. The SHAP framework enables feature attributions such as Shapley values to be visualized as forces where each feature value is a "force" that either increases or decreases the predicted probabilities. In the context of our DL-based NIDS, SHAP local explanation can be used to determine which features of a specific dataset increase or reduce the likelihood of an attack. Cyber-security experts can then use this information to develop cyber-resilient intrusion detection systems.

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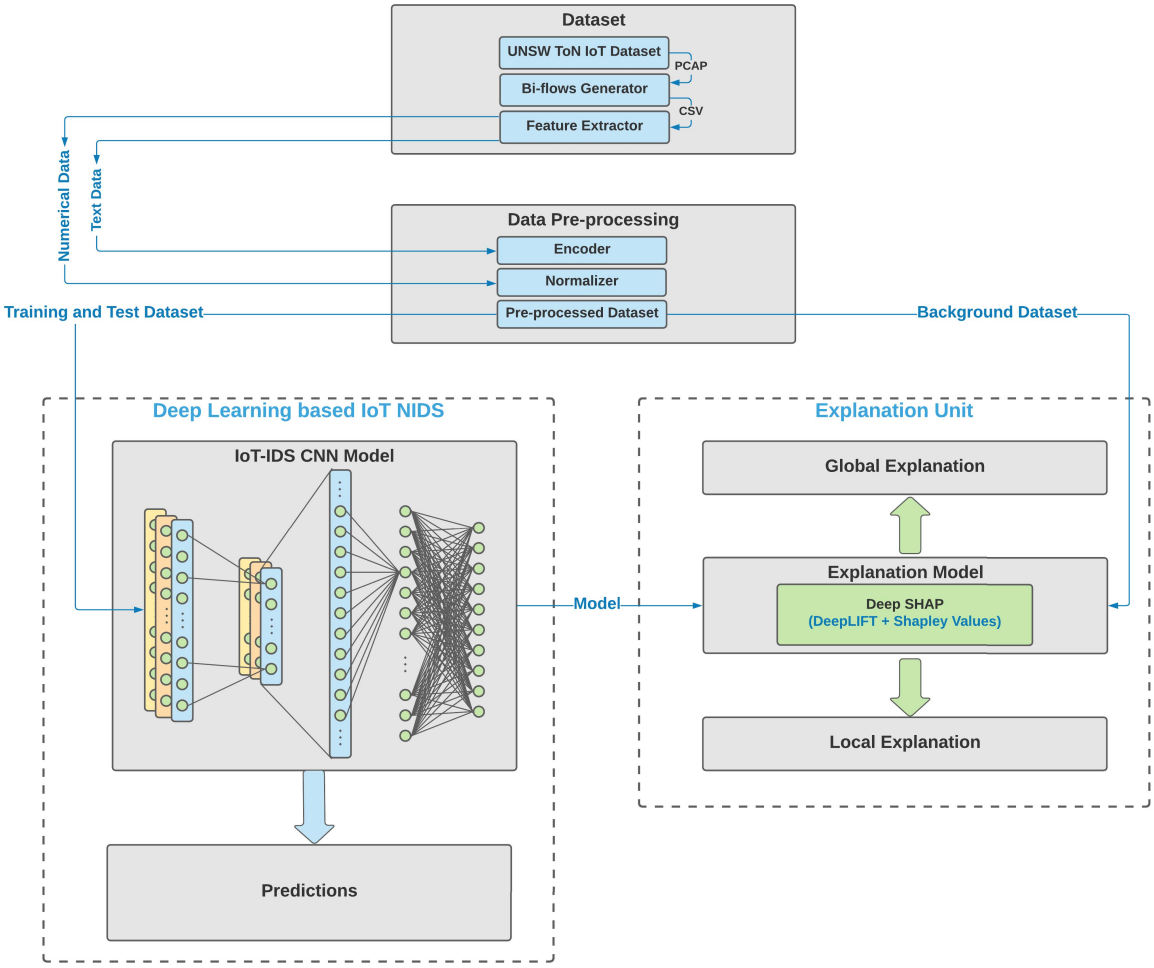


Fig. 4. Proposed explainable-ai framework for intrusion detection in IoT systems.

C. Global Explanation

Global explanation or interpretation is one of the benefits of the SHAP framework. It provides an overview of features that are most important for a model. To explain an entire model, Lundberg and Lee [29] introduced the idea of combining Shapley values in a matrix that has one row per data instance and one column per feature. Several visualization options are available as part of the *shap* Python package for global explanations. Two of the commonly used global interpretation visualizations include in the *shap* Python package are *Summary Plots* and *Dependence Plots*.

SHAP summary plots provide a global summary of the distribution of SHAP values' overall features in the dataset. They show the distribution of feature importance for each feature. Dependence plots, on the other hand, are referred to as the simplest global interpretation plots [28]. Dependence plots show the relationship between the value of a feature and the corresponding Shapley values for each instance in the dataset. While summary plots show the relationship between a feature value and its impact on the prediction, the SHAP dependence plots show the exact form of the relationship.

D. Cyber-Resilience

Intrusion detection is considered a powerful means of achieving cyber-resilience in IoV networks [9]. Cyber-resilience is the ability of a system to absorb disruption or attack, recover to a pre-disruption functionality, and adapt its protocol and configuration to become more resilient to known threats. The proposed explainable DL-based IDS can help achieve cyber-resilience in transportation systems by acting as the first line of defence to detect malicious behaviours in IoV networks. Furthermore, insights derived from the local and global explanations can be used by experts and system designers to adapt the protocol and configuration of IoV systems, so they are more resilient to known threats.

IV. Experiments and Results

This section discusses the details of our experiments and analyses the results. Specifically, this section is divided into three parts: *(1) Experimental Methodology*, *(2) Performance evaluation and analysis*, and *(3) Model Explanations*.

A. Experimental Methodology

1) Overview of Dataset: The experiments in this work were carried out using the ToN_IoT network datasets [37].

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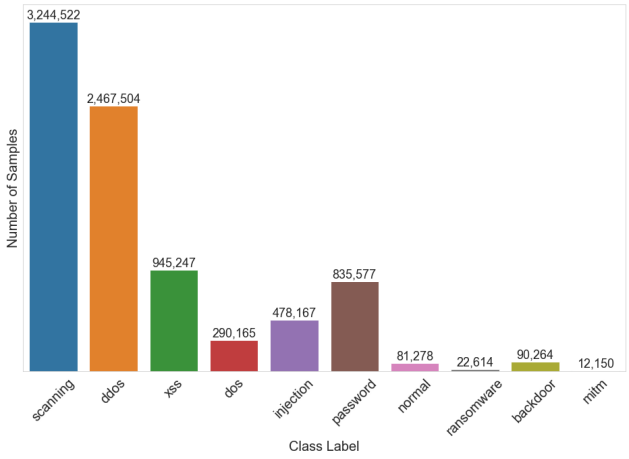


Fig. 5. Composition of the ToN_IoT dataset used for the experiments.

The ToN_IoT datasets are a collection of heterogeneous datasets that contain the actual characteristics of real-world IoT and Industrial IoT networks for evaluating AI and cyber-security systems in applications such as intrusion detection, adversarial machine learning and threat intelligence. In our experiments, we used the raw network datasets in packet capture (PCAP) formats which comprise 'normal' network packet captures and packet captures collected during nine different simulated attacks: Injection, Man-in-the-middle (MITM), Backdoor, Distributed Denial of Service (DDoS), Denial of Service (DoS), Ransomware, Scanning, Cross-site scripting (XSS), and Password attack. The composition of the dataset we used in our experiments is shown in Fig. 5.

2) Flows Generation and Feature Extraction: We used the CICFlowMeter tool [38] to generate bi-directional network flows from the ToN_IoT PCAP files and extract features from the flows. According to Lopez *et al.* [39], the flow-based categorization of traffic helps detect anomalies in network traffic. A total of 83 features (including statistical time-related features) were extracted from the flows. Based on ground truth, each flow was labelled as either 'normal' or using the specific attack name. A brief description of each of the statistical time-related features is provided in Table III.

3) Data Pre-Processing: In this experiment, we found that features such as the source IP, destination IP and source port were either highly correlated or had a minimal contribution to the model output; hence these features were dropped from the dataset. The only non-numerical feature in the dataset is the 'label', and we used the Scikit-learn label encoder to convert the text data to binary. For ease of training, we also normalized all the time-related features to a common scale in the range [0, 1].

4) CNN-Based IDS Architecture: The models used in our experiments are based on the Convolutional Neural Network (CNN) architecture using the ToN_IoT dataset. Convolutional Neural Networks (CNN) is a class of discriminative deep neural networks that have been widely used in many machine learning applications such as computer vision and speech processing. Convolutional neural networks are effective at automatically extracting complex features from heterogeneous data while adapting to changing environments [40] constantly. This characteristic, coupled with the high detection rates and accuracy of CNN-based IDS models, is one reason we propose the CNN architecture in our framework.

As with many deep networks, a CNN architecture consists of an input layer, multiple hidden layers and an output layer. Typically, the hidden layers comprise convolutional, pooling, and fully connected layers. The convolutional layer applies a convolution operation to the input data to abstract it to a feature map. On the other hand, the pooling layer operates on the input feature map blocks to reduce its dimension while retaining the essential information. Finally, the fully-connected layer is simply a multi-layer perceptron neural network where each neuron in its previous layer is connected to every neuron in the next layer. The output layer uses a softmax activation function for classification. A detailed discussion on CNN can be found in [41].

We trained 1D convolutional neural networks for binary and multi-class classification. The binary class IDS model classifies an input network traffic flow as either normal flow or an attack flow, while the multi-class IDS model predicts the actual class of an attack from nine possible attack types.

Implementation: The IDS architecture was implemented using the Keras Python library, which provides a simple API for building deep neural networks in code. The experiments were performed on the Jupyter Notebook using a 2.2GHz 6-Core Intel Core i7 CPU with 32GB RAM. Each model was trained on the CPU using the Adam optimizer with a learning rate of 0.001 over 20 epochs.

A summary of the layers in our binary class and multi-class CNN-based IDS models are shown in Tables IV and V respectively.

B. Model Performance Evaluation and Analysis

To evaluate the performance of our CNN-based IDS models trained on the ToN_IoT dataset, we use four well-known metrics: Accuracy, Precision, Recall and F1-score. To compare the results of our IDS model with the Ground Truth, we use terms such as True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

1) Accuracy: The Accuracy (ACC) metric is used to evaluate the overall performance of a model. It is the ratio of correct predictions to the total number of predictions.

In the case of the binary classification IDS model, ACC was estimated in terms of the true/false positives and true/false negatives as follows:

$$\text{Accuracy (ACC)} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{10}$$

2) Precision: In the context of our IDS, Precision is expressed as the ratio of the number of instances correctly predicted as an attack to the number of instances classified as attack. Formally,

$$\text{Precision (P)} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

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TABLE III
EXTRACTED FEATURES AND THEIR DESCRIPTIONS

Feature Name	**Description**
flow_duration	Duration of the flow in Microsecond
tot_fwd_pkts	Total packets in the forward direction
tot_bwd_pkts	Total packets in the backward direction
totlen_fwd_pkts	Total size of packet in forward direction
totlen_bwd_pkts	Total size of packet in backward direction
fwd_pkt_len_min	Minimum size of packet in forward direction
fwd_pkt_len_max	Maximum size of packet in forward direction
fwd_pkt_len_mean	Mean size of packet in forward direction
fwd_pkt_len_std	Standard deviation size of packet in forward direction
bwd_pkt_len_min	Minimum size of packet in backward direction
bwd_pkt_len_max	Maximum size of packet in backward direction
bwd_pkt_len_mean	Mean size of packet in backward direction
bwd_pkt_len_std	Standard deviation size of packet in backward direction
flow_byts_s	Number of flow bytes per second
flow_pkts_s	Number of flow packets per second
flow_iat_mean	Mean time between two packets sent in the flow
flow_iat_std	Standard deviation time between two packets sent in the flow
flow_iat_max	Maximum time between two packets sent in the flow
flow_iat_min	Minimum time between two packets sent in the flow
fwd_iat_min	Minimum time between two packets sent in the forward direction
fwd_iat_max	Maximum time between two packets sent in the forward direction
fwd_iat_mean	Mean time between two packets sent in the forward direction
fwd_iat_std	Standard deviation time between two packets sent in the forward direction
fwd_iat_tot	Total time between two packets sent in the forward direction
bwd_iat_min	Minimum time between two packets sent in the backward direction
bwd_iat_max	Maximum time between two packets sent in the backward direction
bwd_iat_mean	Mean time between two packets sent in the backward direction
bwd_iat_std	Standard deviation time between two packets sent in the backward direction
bwd_iat_tot	Total time between two packets sent in the backward direction
fwd_psh_flags	Number of times the PSH flag was set in packets travelling in the forward direction (0 for UDP)
bwd_psh_flags	Number of times the PSH flag was set in packets travelling in the backward direction (0 for UDP)
fwd_urg_flags	Number of times the URG flag was set in packets travelling in the forward direction (0 for UDP)
bwd_urg_flags	Number of times the URG flag was set in packets travelling in the backward direction (0 for UDP)
fwd_header_len	Total bytes used for headers in the forward direction
bwd_header_len	Total bytes used for headers in the backward direction
fwd_pkts_s	Number of forward packets per second
bwd_pkts_s	Number of backward packets per second
pkt_len_min	Minimum length of a packet
pkt_len_max	Maximum length of a packet
pkt_len_mean	Mean length of a packet
pkt_len_std	Standard deviation length of a packet
pkt_len_var	Variance length of a packet
fin_flag_cnt	Number of packets with FIN
syn_flag_cnt	Number of packets with SYN
rst_flag_cnt	Number of packets with RST
psh_flag_cnt	Number of packets with PUSH
ack_flag_cnt	Number of packets with ACK
urg_flag_cnt	Number of packets with URG
cwe_flag_count	Number of packets with CWR
ece_flag_cnt	Number of packets with ECE
down_up_ratio	Download and upload ratio
pkt_size_avg	Average size of packet
fwd_seg_size_avg	Average size observed in the forward direction
bwd_seg_size_avg	Average size observed in the backward direction
fwd_byts_b_avg	Average number of bytes bulk rate in the forward direction
fwd_pkts_b_avg	Average number of packets bulk rate in the forward direction
fwd_blk_rate_avg	Average number of bulk rate in the forward direction
bwd_byts_b_avg	Average number of bytes bulk rate in the backward direction
bwd_pkts_b_avg	Average number of packets bulk rate in the backward direction
bwd_blk_rate_avg	Average number of bulk rate in the backward direction
subflow_fwd_pkts	The average number of packets in a sub flow in the forward direction
subflow_fwd_byts	The average number of bytes in a sub flow in the forward direction
subflow_bwd_pkts	The average number of packets in a sub flow in the backward direction
subflow_bwd_byts	The average number of bytes in a sub flow in the backward direction
init_fwd_win_byts	The total number of bytes sent in initial window in the forward direction
init_bwd_win_byts	The total number of bytes sent in initial window in the backward direction
fwd_act_data_pkts	Count of packets with at least 1 byte of TCP data payload in the forward direction
fwd_seg_size_min	Minimum segment size observed in the forward direction
active_min	Minimum time a flow was active before becoming idle
active_mean	Mean time a flow was active before becoming idle

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TABLE III
(Continued.) EXTRACTED FEATURES AND THEIR DESCRIPTIONS

active_max	Maximum time a flow was active before becoming idle
active_std	Standard deviation time a flow was active before becoming idle
idle_min	Minimum time a flow was idle before becoming active
idle_mean	Mean time a flow was idle before becoming active
idle_max	Maximum time a flow was idle before becoming active
idle_std	Standard deviation time a flow was idle before becoming active

TABLE IV
BINARY-CLASS CNN-BASED IDS ARCHITECTURE

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 79, 64)	256
conv1d_1 (Conv1D)	(None, 77, 64)	12352
max_pooling1d (MaxPooling1D)	(None, 38, 64)	0
dropout (Dropout)	(None, 38, 64)	0
flatten (Flatten)	(None, 2432)	0
dense (Dense)	(None, 64)	155712
dropout_1 (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 2)	130

TABLE V
MULTI-CLASS CNN-BASED IDS ARCHITECTURE

Layer (Type)	Output Shape	Param #
conv1d (Conv1D)	(None, 79, 64)	256
conv1d_1 (Conv1D)	(None, 77, 64)	12352
max_pooling1d (MaxPooling1D)	(None, 38, 64)	0
dropout (Dropout)	(None, 38, 64)	0
flatten (Flatten)	(None, 2432)	0
dense (Dense)	(None, 64)	155712
dropout_1 (Dropout)	(None, 64)	0
dense_1 (Dense)	(None, 10)	650

3) Recall: Recall is the ratio of the number of instances correctly predicted as an attack to the total number of attack instances. Formally,

$$\text{Recall (R)} = \frac{\text{TP}}{\text{TP + FN}} \tag{12}$$

4) F1-Score: The F1 score is expressed mathematically as the harmonic mean of precision (P) and recall (R):

$$\text{F1 score} = 2 \cdot \frac{\text{P} \cdot \text{R}}{\text{P + R}} \tag{13}$$

While our experiments aim to interpret the output of the two IDS classifiers trained on the ToN_IoT dataset, it was still important to ensure that the classifiers had reasonable detection rates. The overall performance of the IDS models is shown in Table VI.

In Table VII, we provide a performance comparison between our CNN-based IDS model and other state-of-the-art models that have been trained on publicly available IoT datasets. Our CNN-based IDS achieved an accuracy of 99.15%, precision of 99.10%, recall of 99.15% and F1 score of 98.83%. Based on these results, we can easily conclude that CNN-based IDS is appropriate for detecting cyber-attacks in IoV networks.

TABLE VI
CNN-BASED IDS CLASSIFICATION RESULTS

IDS Model	Accuracy %	Precision %	Recall %	F1 Score %
Binary-class ToN_IoT IDS model	99.15	99.10	99.15	98.83
Multi-class ToN_IoT IDS model	90.55	90.75	90.55	90.22

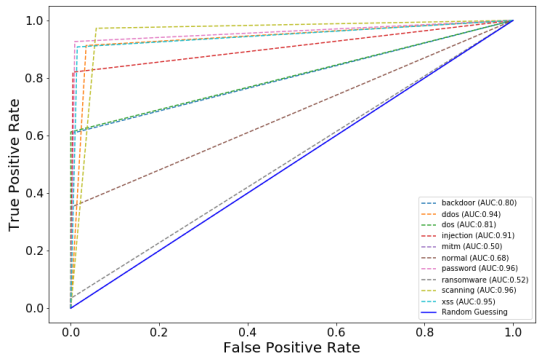


Fig. 6. ROC curve and AUC.

5) Receiver Operating Characteristics (ROC) Curve: The ROC curve is an evaluation metric typically used in many binary classification problems and is being extended to multi-class problems. On the multi-class ROC curve, the true positive rate is plotted as a function of the false positive rate for each class label. In this work, we use the ROC curve in Figure 6 to show how well our model distinguishes between the various attack and normal traffic classes. We show that the scanning and password attack classes have the highest Area Under Curve (AUC) score, which shows that the model performs well at predicting scanning and password attacks compared to other attack types and normal traffic.

C. Model Explanations

In this section, we describe how the Deep SHAP explanation algorithm can be used to explain the predictions of our CNN-based IDS model. As discussed in Section III, the Deep SHAP algorithm is a fast model-specific approximation method for providing both local and global explanations for the output of deep neural networks. The authors of the SHAP algorithm developed the Python *SHAP* package

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TABLE VII
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART MODELS

Model	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Fedhealth [42]	UCI Smartphone dataset	99.13	99.03	96.85	97.90
FLIDS [43]	MIMIC dataset	98.17	98.79	96.45	97.57
DeepFed [44]	Real CPS dataset	99.20	98.86	97.36	98.10
XSRU-IoMT [45]	ToN_IoT dataset	99.38	99.39	98.99	99.37
Proposed CNN-based IDS	**ToN_IoT dataset**	**99.15**	**99.10**	**99.15**	**98.83**

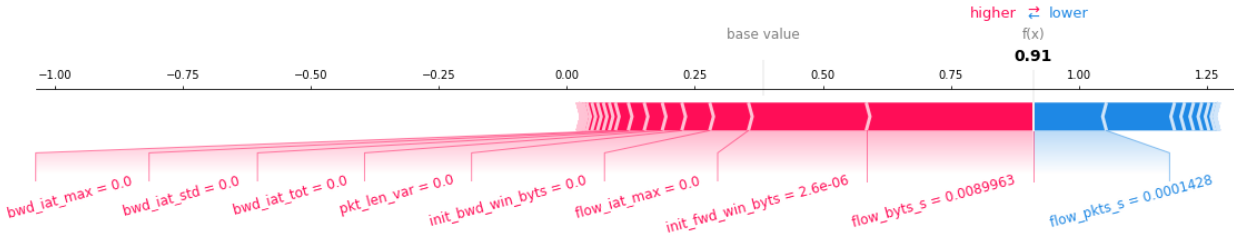


Fig. 7. Explanation of the multi-class CNN-based IDS on scanning attack.

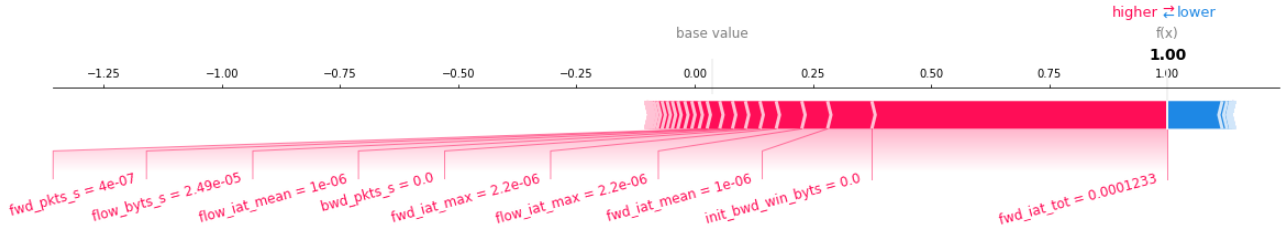


Fig. 8. Explanation of the multi-class CNN-based IDS on DoS Attack.

(https://github.com/slundberg/shap), which comes with different visualization plots that enable local and global interpretation from estimated SHAP values. Using this Python package, we can explain the predictions of our CNN-based IDS for single data instances using the *SHAP Force Plots*.

A *Force Plot* helps to visualize feature attributions of individual instances, where each feature value represents a force that either supports or contradicts a prediction. In this work, we use the *Force Plots* to visualize the packet flow features that supports or contradicts the prediction of our IDS model. *Force Plots* also show the size or influence each feature has on the prediction. In Figures 7 and 8, we show visualizations of the interpretation of our multi-class CNN-based IDS on two (2) flow instances representing a scanning attack and a dos attack, respectively. For the first instance (Fig. 7), the model's prediction probability is shown in bold as 0.91, which implies that our multi-class IDS model is 91% certain that the flow is a scanning attack.

The numbers on the force plot (stacked arrows) are values of the features for the instance or network flow being predicted. The red arrows (and corresponding features) increase the probability that our multi-class IDS model predicts the flow as a scanning attack, while the blue arrows lower the probability of a scanning attack prediction. It can also be seen that the feature 'Number of Flow Bytes per sec (flow_byts_s)' has the most substantial influence on the prediction due to it having

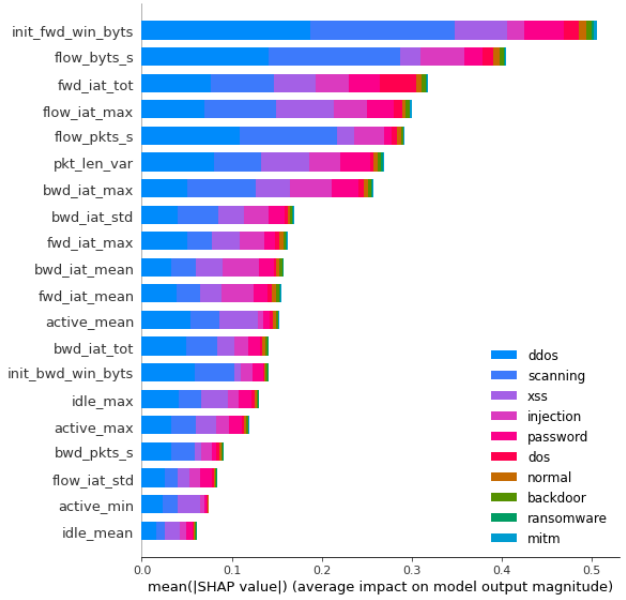
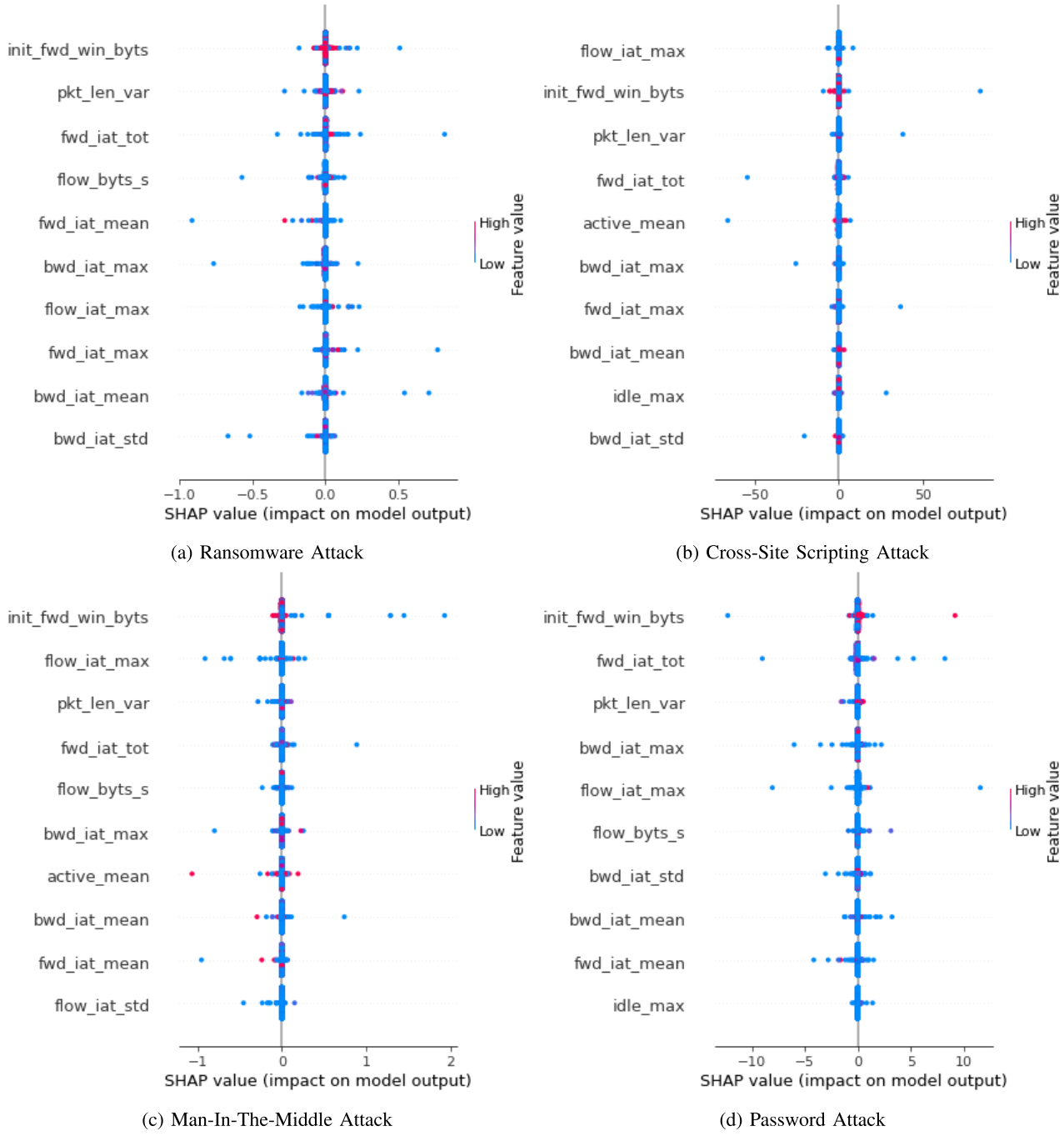


Fig. 9. SHAP feature importance measured as the mean absolute Shapley values.

the largest magnitude. Typically, a high number of flow bytes per second is one of the characteristics of scanning attacks (as shown in Table II).This type of information provided by

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(a) Ransomware Attack

(b) Cross-Site Scripting Attack

(c) Man-In-The-Middle Attack

(d) Password Attack

Fig. 10. Top 10 features contributing to the prediction of Ransomware, cross-site scripting (XSS), man-in-the-middle and password attacks.

the explanation unit helps cybersecurity experts and other decision-makers to be more convinced of the threat detection capability of an IDS. The model probability of 1 for the prediction visualized in Fig. 8 indicates a 100% certainty from our multi-class model that the instance is a dos attack. In this case, however, the feature 'Total time between two packets sent in the forward direction (fwd_iat_tot)' has the strongest influence on the prediction of the instance as a dos attack.

In Fig. 9, we show the *SHAP Importance Plot* for Shapley values of the first 5000 instances of our test dataset. In this plot, the features with large absolute Shapley values are classed as important as they have a higher average impact on the model output. To provide visualization of the features in our dataset that are most important for the proposed multi-class IDS model to predict a particular attack class, we use the *SHAP Summary Plot*. In contrast to the *Force Plots*, the *SHAP Summary Plots* is a visualization plot for global interpretation that communicates global feature importance for the entire model with the most important features on top and least important features at the bottom. In Fig. 10, we show visualisation of the top 10 flow features that contribute to the multi-class IDS prediction of a *Ransomware*, *Cross-Site Scripting*, *Man-In-The-Middle* and *Password* attacks. Each of the plots displays an information-dense summary of the relationship between the

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value of a feature in our dataset and its impact on the output of our multi-class IDS model. Each point on the summary plot corresponds to a Shapley value for a feature and an instance. The colour of the points in the summary plot represents the value of the features, where 'blue' indicates a low feature value while 'red' indicates a high feature value. The main difference between the Summary plot (Fig. 10) and the importance plot (Fig. 9) is that the Summary plot shows both the negative and positive effects of each of the features on the model output.

V. Conclusion and Future Work

In this work, we have proposed an explainable deep learning framework that helps to improve the transparency and resilience of deep learning-based intrusion detection systems used in IoT networks such as IoV ones. Based on this framework, cybersecurity experts can develop more resilient intrusion detection systems that can absorb an attack, recover to a pre-disruption functionality, and adapt its protocol and configuration to become more resilient to known threats. The proposed framework uses the SHapley Additive exPlanations (SHAP) method to provide local and global explanations for any deep learning-based IDS. The SHAP method has solid theoretical foundations in cooperative game theory and has been used widely in many computer vision applications to explain the outputs of deep learning models. To validate our proposed framework, we used the ToN_IoT dataset to train a convolutional neural network and then used the DeepSHAP to interpret the network's predictions. However, a drawback to using SHAP is that it is computationally intensive and expensive to run. It has also been found to be vulnerable to adversarial attacks. In future work, we will investigate the extent to which SHAP is vulnerable to malicious attacks and seek to explore existing or new defences that can be used to improve its robustness in IoT systems.

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Ayodeji Oseni is a doctoral research student with the School of Engineering and Information Technology, University of New South Wales (UNSW) Canberra, Australia. His research interests focuses on the intersection of security, privacy, and machine learning within the context of the IoT ecosystems.



Nour Moustafa (Senior Member, IEEE) received the bachelor's and master's degrees in computer science from the Faculty of Computer and Information, Helwan University, Egypt, in 2009 and 2014, respectively, and the Ph.D. degree in cyber security from the University of New South Wales (UNSW) Canberra, Canberra, Australia, in 2017. He was a Post-Doctoral Fellow with UNSW Canberra from June 2017 to December 2018, where he is the Leader of intelligent security with SEIT. His areas of interests include cyber security, in particular, networks security, the IoT security, intrusion detection systems, statistics, deep learning, and machine learning techniques.



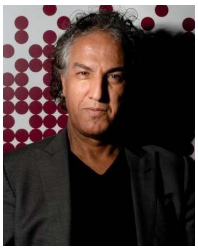
Gideon Creech is an Independent Researcher with Volos Services Pty Ltd., Wamboin, NSW, Australia. His research interests include cyber warfare, hacking, intrusion detection systems, and computer security.



Nasrin Sohrabi received the Ph.D. degree in computer science from RMIT University, Australia. She is currently a Post-Doctoral Research Fellow in cloud, systems, and security discipline with the School of Computing Technologies, RMIT University. Her expertise is in the areas of scalability and performance of distributed systems, consensus algorithms, and the systems' security.



Andrew Strelzoff was the Head of the Computer Science and Graduate Program at the School of Computing, The University of Southern Mississippi, before leaving in 2012 to work at the Information Technology Laboratory (ITL). He is a Data Scientist with ITL, U.S. Army Engineer Research and Development Center (ERDC), United States Army Corps of Engineers. As the Lead Scientist for the Engineering Resilient Systems Program at ITL. He helped the program grow to a full ITL business area employing more than 100 full time employees. Currently, he is working as a senior technical expert in area of applications of artificial intelligence to risk and resilience under the direction of ERDC senior technical staff Dr. Igor Linkov.



Zahir Tari is a Full Professor with RMIT University, Australia; and the Research Director of the RMIT Cyber Security Research and Innovation (CCSRI). His expertise is in the areas of system's performance (e.g. P2P, cloud, and edge/the IoT) as well as system's security (e.g. SCADA, smartgrid, cloud, and edge/the IoT).



Igor Linkov is the Senior Science and Technology Manager with the U.S. Army Engineer Research and Development Center (ERDC) and an Adjunct Professor with Carnegie Mellon University. He develops methods and tools for measuring risk and resilience in interconnected networks and applies these tools to critical infrastructure, transportation, energy and cyber systems, and supply chains. He has received multiple USACE, army, and DOD awards and civilian service medals, including the 2020 DOD Top Scientist Award.

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